PORTFOLIO ANALYTICS

How to read this page

A companion to the page: what each part shows, in plain words, for a reader new to investing.

Mason Bennett · M.S. Finance · University of Arkansas

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Start here

The page takes a handful of investments you choose and looks at how their prices moved. Then it works out how different ways of splitting money between them would have done.

You do not need this guide to use it. Every chart says what it shows, and every figure explains itself one tap away. Read on when you want the why, or a word you have not met.

Names in bold, like **Tickers** or **Performance**, are written as the page prints them, so you can find them on the screen. The glossary at the end explains the rest.

Every figure describes the past: the prices between two dates you pick. Nothing is a forecast or a recommendation. Nothing you do is saved anywhere but in your own browser.

Fitted, then tested

The page asks one question: how should the money have been split? A split is a portfolio, or a mix: a set of weights saying what share of the money goes in each investment.

Mixes are scored with the Sharpe ratio. It is the return earned above a safe rate, for each unit of risk taken. Risk here means how much the value swung up and down, which the page calls volatility. A higher Sharpe ratio is better.

Three mixes recur. The equal split (equal weight) puts the same share in everything; nobody looked at the prices to choose it. The lowest-risk mix (GMV, global minimum variance) is the one that swung least. The best-tradeoff mix (tangency) has the highest Sharpe ratio.

The last two are fitted: chosen by looking at the prices, then scored on those same prices. That is in-sample, like setting an exam after reading the answers. It shows the best split with hindsight, not what anyone could have earned.

The fair test is out-of-sample. Choose the weights on earlier prices, then hold them, unchanged, over later days they have not seen. The **Does it hold up?** chapter runs that test, and the Overview quotes it as published.

The gap can be large. Take the **Five mega-caps** set, shares in some of the largest US companies. Fitted to its prices up to Sep 2026, at a 2.0% safe rate, the best-tradeoff mix scored 1.107. Held over years it had not seen, it earned 0.659, below the 0.864 of the equal split. On the **Cross-asset** set, funds holding stocks, bonds, gold and property, it went the other way: 0.883 out of sample, against 0.704.

So read an in-sample figure as a description of the past. The published out-of-sample result is the fairer test, and it came out both ways. It is not a promise about the years ahead.

The first screen

The masthead

At the top: the title, who made it and a link to this guide. Just below sits the bar holding your basket and the settings, described in the next section.

On a first visit the page opens on a saved example, the cross-asset set, so it is never blank.

The question and the answer

The page opens on Overview, with one question in large type: **Can a smarter mix beat an equal split?** Two sentences answer it.

The first is about your tickers, with hindsight: what the starting amount became in the best-tradeoff mix, chosen on these very prices, against the equal split.

The second is Mason's published test, without hindsight. If your tickers are exactly one of the three published sets, it quotes that set's figures. On any other basket it says what the test found and offers to run it on yours.

A word with a dotted underline, such as **With hindsight**, opens a short definition when you tap it.

The two cards

The first card draws what the starting amount became in the three mixes and the benchmark, each line named at its end.

The second, **Hindsight flatters the optimizer** on the example, sets each mix's Sharpe ratio in Mason's published test fitted with hindsight on the test's whole window (a hollow bar) beside its figure on years it had not seen (a solid bar), the same pairs Does it hold up? draws. On a basket that was never published, your bars stand alone, with hindsight on this window, and the five mega-caps follow as the reference, under their own name.

Every card has two folds. **Show the numbers** opens the figures behind the picture; **How it is calculated** opens the method.

The second card's numbers open on the highest Sharpe ratio your tickers reached, then four more: **Equal-Weight Sharpe**; **Tangency Sharpe (in-sample)**, with a plus-or-minus standard error, a rough sense of how far luck alone could move it; **GMV Sharpe (in-sample)**; and the benchmark's, such as **S&P 500 Sharpe.** The small i beside each explains it. Under them is the published sentence as masonjbennett.com prints it. **Walk-forward test** replays that test, and the last link opens the method note.

Why, and what to try next

Three short facts say why the answer comes out as it does: how much of the best-tradeoff mix sits in one holding, how far one published weight swung with the years it looked back over, and what the published one-year tests found.

Try another basket: loads a published set in one tap, or type your own tickers in the bar. The last line says who built the page and links its source code, the method note and this guide.

The settings

The settings sit in the basket bar across the top, which stays in view as you scroll: the tickers, the ready-made baskets and the dates, then **Assumptions**, which opens the rest in a strip under the bar.

Most changes apply at once, with no button to press. New tickers apply when you press Enter. Only a new basket, new dates or a new benchmark fetch prices.

At the bar's right end, **Prices to** says which day's closing prices the figures run to. While the saved example shows, the word **example** sits beside the date until fresh prices arrive.

On a phone the bar folds into one row at the top that says what is set. Tap **Settings** on it to open every setting in a sheet.

Tickers

A ticker is the short code a market uses for an investment: AAPL for Apple, VTI for a fund holding the whole US stock market. Each ticker is a chip on the bar, and its cross takes it out.

+ Add opens a box with the whole list. Type between 3 and 10 codes, separated by commas, and press Enter. The codes are Yahoo Finance's, so funds and indexes work too.

Baskets

Baskets fills the list with a ready-made basket. The first three, marked **Published**, are the sets whose out-of-sample result is published. Here they are recomputed in-sample on the prices loaded, so their figures are not the published ones. **More baskets** opens the rest.

Date range

The dates on the bar open **Start** and **End**: the window, the stretch of prices every figure comes from. The end is today unless you pick a day; **End today instead** puts it back. A new window fetches prices again.

Risk-free rate

The risk-free rate is what a safe investment paid over the same time: short-term US Treasury bills. A Sharpe ratio counts only the return above it. By default it is the average yield over your window, from FRED, the Federal Reserve Bank of St. Louis's public data.

Use today's rate instead swaps in the latest yield. Typing a rate fixes it there; **Use the rate over the window** goes back. The note under the field always says which rate is in use.

Benchmark

The benchmark is what everything is measured against: a broad market index, or a fund that tracks one. The choices are S&P 500 (Recommended), Nasdaq 100, Dow Jones, Russell 2000, MSCI World (URTH) and Total Market (VTI).

Short selling

Off, every weight is zero or more: money only goes into an asset. With **Allow short positions** on, a weight may go below zero. That is a short position: you borrow an asset and sell it, betting its price will fall.

The money from that sale goes into the other assets, so their weights alone can add to more than the whole. All the weights together still make the whole. The note under the switch gives the limits. Switching solves the mixes again on the prices already loaded.

Starting amount

The sum every growth chart starts from. It changes the dollar figures only, never a return or a ratio. The growth charts carry the same field, and a change in either place moves both. It is kept in this browser and never travels in a shared link.

The chapters

The row of five chapters above the question holds the analysis, in reading order: **Overview**, **Performance**, **Diversification**, **Build a portfolio** and **Does it hold up?**. Everything in them comes from the window and the tickers in the settings.

Each chapter after the first opens on one sentence in large type that says what it found, then the chart behind it. Below that, every further analysis is a row: a title and its finding on one line. A row opens in place to its chart or table, and stays open until you reload.

Beside the three mixes there is your own, which the page calls **Custom** and you set in **Build a portfolio**. Where a table has a column per mix, the columns use the finance names: Equal-Weight, GMV, Tangency, Custom.

Overview

Where should I start?

The chapter the page opens on: the question, the answer, two cards and the baskets to try, described above.

Open Overview

Performance

How did each investment and each mix grow, and how far did they fall?

It opens on where a starting amount ended highest and lowest, and one chart of what it became. The rows on the returns come first; the rows on the falls follow, under How deep the falls went.

The returns

How did each investment grow, and how bumpy was the ride?

The chart shows where a starting amount ended. **Each asset** draws one line per investment and the benchmark; **The mixes** draws the three mixes, your own once you have set one in Build a portfolio, and the benchmark.

The buttons above the asset lines switch them on and off. The box after **Growth of $** is the starting amount, the same one as in the settings. **Show the numbers** gives each line's end figures, and **How it is calculated** says how.

Calendar years

Each calendar year's return for the four mixes and the benchmark.

Month by month

Months of picks a mix for a grid of every month, shaded from loss to gain. The same months are a table under **Show the numbers**.

Summary Statistics

One row per asset, and a last row for the benchmark. It gives the average yearly return and the volatility, both as yearly percentages. Volatility is not a gain or a loss: the higher it is, the further the value tended to stray from its average, either way.

Then the worst and best single days, and two measures of shape. Skewness says whether the biggest moves leaned toward gains or toward losses. Excess kurtosis above zero means extreme days came more often than a bell curve would predict.

Return Distribution

Pick an **Asset** to see its daily returns two ways. **Histogram + Normal Fit** counts how often each size of move happened, beside the bell curve of a normal distribution. The far ends are the tails: the rare big days up and down. The title says whether they were fatter than the curve's, which means big days came more often than it allows.

Q-Q Plot lines the daily returns up from worst to best, each against where the bell curve would put it; the page calls those places quantiles. Returns that matched the curve would sit on the straight line. Dots that bend away at the ends are days more extreme than the curve allows.

How deep the falls went

How far did things fall, and how long did the falls last?

Performance's second list of rows. The Drawdowns row names the asset that fell furthest from a high, and when; its chart opens on that asset.

The portfolios in a fall

Pick a mix to list its **Deepest falls** in your window: from which high, to which low, how deep, and how long it took to get back. **Named falls in this window** sets every mix against the same well-known market falls, over the same days.

Drawdowns

A drawdown is how far an investment stands below its highest price so far. The chart follows one asset, picked above it, day by day. **Worst drawdown by asset** lists each asset's deepest, with its dates.

Rolling volatility

Volatility over a moving window of recent trading days, so calm and stormy stretches show as low and high lines. The buttons above it set the window's length.

Risk-adjusted metrics

Each asset's Sharpe ratio and Sortino ratio, and the benchmark's. The Sortino ratio is the same idea counting only the swings below the safe rate, since few investors mind the swings up.

CAPM beta and alpha

CAPM, the capital asset pricing model, is the textbook idea that part of an asset's return simply follows the market. Beta says how much an asset tended to move when the benchmark moved: above one, more; below one, less. Alpha is the yearly return left over that the benchmark's moves do not explain.

Both describe this window, not a skill that will carry on. The dashed line at one is the benchmark's own beta.

Open Performance

Diversification

Which of the investments moved together?

One part: how closely each pair moved.

How they move together

Do these investments rise and fall together?

Correlation runs from minus one (always moving opposite) through zero (unrelated) to plus one (always together). Investments that do not move together are what makes spreading money between them worth anything.

The chapter opens on the most and the least correlated pair, and a shaded grid of every pair; the figures are under **Show the numbers**. Three rows follow.

Rolling pairwise correlation

Pick two assets and a window length to see how their correlation changed over time, not one figure for the whole window.

Equal weights against each on its own

What the equal split did in your window: how much it swung in a year, against how much the assets swung on average, one by one. The gap is what diversification took away.

Daily covariance matrix

The raw material the optimizer works from: a grid with a row and a column per asset, saying how each pair's daily returns moved together. Where an asset meets itself, the figure is how much it swings on its own.

The figures are tiny and hard to read by eye. Correlation divides each by how much the two assets swing, which makes it easier to read but drops the size of the swings. The optimizer needs the covariances for exactly that.

Open Diversification

Build a portfolio

Which mix would have been best, and how does your own compare?

It leads with the frontier and your own mix beside it. The tables of the fitted mixes follow as rows.

The mixes

Which mixes of these assets did best, with hindsight, and where does your own sit?

It opens on the best-tradeoff mix's weights and its in-sample Sharpe ratio, against the equal split's. Everything fitted here is chosen on the same prices it is scored on.

Then the efficient frontier. Volatility runs across and return up, so a point further left swung less and a point higher earned more. Each point on the curve is the least volatile mix for its return; the lowest-risk mix sits at the far left. A straight line from the risk-free rate just touches the curve at the best-tradeoff mix, which is why its finance name is tangency.

The single assets, the even split and your own mix, **Custom**, are marked too. The line under the chart's title says how far your mix sits from the curve.

Beside it is **Your mix (Custom)**: a slider and a box per asset, each starting equal. The page divides every setting by their total, so the shares always make the whole; raising one lowers every other. The point on the frontier and the five figures under the sliders follow as you move, and **Reset to equal weights** appears once you have.

Show the numbers gives your shares after that division, the ones every figure of your mix uses. Your weights travel in the page's link. Your mix is a column of the scorecard below and a line on Performance's growth chart.

Scorecard

Every mix in one table, a column each: **Equal-Weight**, **GMV**, **Tangency**, your mix as **Custom**, and the benchmark last. Its rows give return, risk, the falls, how the mix moved with the benchmark, and how fragile its weights are. The main ones are return, volatility, the Sharpe and Sortino ratios and the maximum drawdown, its deepest fall.

The two fitted columns say **weights chosen on this window**; under How it is calculated, **The walk-forward test** opens what such weights earned on later years. The buttons above add other ways of building a mix, such as **Risk parity**. **Show every row** shows the rows left out at first, and **What each row means** explains each.

Beside the usual downloads, **Download Excel, with formulas** gives a workbook that rebuilds most of the scorecard as spreadsheet formulas from the closing prices. The fitted weights go in as plain numbers, and a few rows are left out; the workbook's first sheet names them.

What each mix holds

How much of the money each mix puts in each asset, as bars. Under **Show the numbers** the same weights are a table, with your mix and any column added to the scorecard beside them.

If the averages were a little off

The means are the assets' average returns. Their noise is how far those averages, worked out from a limited stretch of prices, could be off by chance. The page redraws the averages many times within that noise and solves the mix again each time, a dot per draw, to show how far its weights could swing. **Redraw** draws a fresh set.

Risk contribution

Pick a mix to compare each asset's share of its money with its share of the return and of the risk. A small weight in a volatile asset can carry a large share of the risk. The table under **Show the numbers** sets every mix's shares side by side.

What if one expected return were different?

This row opens **What-if on these prices**. Pick an asset, then slide its expected return, the average yearly return the optimizer assumes it earns, **up to two standard errors either side of its average**. At each step the best-tradeoff mix is solved again, and the sentence prints its new weight in that asset. The lowest-risk mix never moves, because it does not use expected returns at all.

Why do fitted weights swing so much? The optimizer must estimate, from the prices, each asset's average return, how much it swings and how each pair moves together. The lines under the slider count those estimates, and say how many years of prices it would take to pin one average down. That is usually far more than the window holds, so each estimate is rough, and a small change in one can move the weights a lot.

Open Build a portfolio

Does it hold up?

Would the best mix have done as well without hindsight?

It leads with the published walk-forward test, the project card's sentence under its chart. The rows run the same test on your basket, show each published hold, and show how much the fitted weights move with the years they are fitted on.

The walk-forward test

What did a fitted mix earn on days it had not seen?

This is the out-of-sample test. Weights are chosen on earlier prices and held unchanged for a stretch called a hold. Then they are chosen again on everything up to that point and held again, to the end. Only the returns earned during the holds count.

In-sample against out-of-sample

For each mix, the Sharpe ratio on the prices it was chosen from, beside the one it earned afterwards. The distance between them is what hindsight was worth. The chapter leads with the published test's three sets this way, the project card's sentence and the method note under the chart.

Show the numbers holds the published figures in one table, and **How it is calculated** the rules of the published run.

Your basket: the same test, run now

The test on the tickers in your settings, solved when the row opens. **Fit on** sets what each choice of weights may look at, **All prior history** or only the most recent stretch. **Hold for** sets how long each hold lasts.

Both count in returns. There is one daily return per trading day, so a count of returns is a count of trading days. **Add the four other constructions** adds the scorecard's other ways of building a mix.

Hold by hold

In your basket's row: each hold's dates and the Sharpe ratio each mix earned in it. Under them, **Weights of** picks a fitted mix to show the weights it held in each hold.

Hold by hold, as published · Five mega-caps · Seven sector ETFs · Cross-asset

One part per published set: each hold, and the weights the two fitted mixes held. **Rerun this set on fresh prices** puts that set into your settings, on the published run's dates and rate with short selling off, and opens your basket's row to run the test again on prices fetched now.

If the fitting window changed

How much do the chosen weights depend on which years they were chosen from?

Its rows, under **If the fitting window changed**, fit the lowest-risk mix and the best-tradeoff mix again on several windows. All of them end on the last day; they run from the most recent year back to the whole window.

Weights across windows

The row's line says how far one asset's weight moves between windows. Open it for a bar per window for each asset's weight. The switch above the chart moves between **GMV** and **Tangency**, and **Show the numbers** holds the weights as tables. Weights that jump between windows depend heavily on which years were used.

Portfolio metrics across windows (in-sample)

Each window's mix, scored on that same window's prices.

Your mix across windows (Custom)

Your own mix, held fixed, scored on each window too, beside the two fitted mixes chosen there.

Open Does it hold up?

On every tab

The i marks

A small i beside a figure or a column explains it. Point at it, move to it with the Tab key, or tap it; tap again or press Escape to close it. It explains three ways: in plain words, in finance terms and as the formula.

Under a part's opening sentence, **The formula** opens the formula behind that sentence's figure, with the page's own numbers in it.

Downloads

Every table has **Download CSV** and **Download Excel** under it, for the table as you see it.

The link in the address bar

Once you change something, the address bar holds a link to exactly this view: the tickers, dates, rate, benchmark, short selling, your custom weights and the chapter. Send it and the other person opens the same analysis, on the latest prices unless you set an end date. The starting amount stays behind.

The keyboard

Press Ctrl and K together (Command and K on a Mac) for a small box that opens a chapter, a part of one or this guide as you type its name.

Glossary

Words the page uses

Annualised
Turned into a yearly figure. The page counts a year as 252 trading days.

Basket
The set of tickers in the settings: the investments every figure is about.

Benchmark
What everything is measured against: a broad market index, or a fund that tracks one. No weights are chosen for it; it is there to compare with.

Capital allocation line
On the efficient frontier chart, the straight line from the risk-free rate through the best-tradeoff mix: every split between the safe rate and that mix.

Construction
A rule for building a mix from the prices: the equal split, the lowest-risk mix, the best-tradeoff mix, and the others the scorecard can add.

Correlation
How closely two sets of daily returns move together, from minus one (always opposite) to plus one (always together).

Covariance
How two sets of daily returns move together, including how big their swings were. The optimizer works from these, one for every pair.

Drawdown
How far an investment or a mix stands below its highest value so far.

Efficient frontier
The curve of the least volatile mix for each level of return. A mix below it could have had less risk for the same return.

Equal weight
The equal split: the same share of the money in every asset, kept even each day. Nobody looked at the prices to choose it, which makes it the fair yardstick.

Estimation window
The stretch of prices the weights are chosen from; the page also calls it a lookback. Does it hold up? changes it to show how much it matters.

ETF
An exchange-traded fund: one ticker that holds many investments and is bought and sold like a share. A sector ETF holds the companies of one industry.

Excess kurtosis
A measure of how often extreme days happen. Zero for a bell curve; above zero, big moves came more often than a bell curve allows.

Expected return
The average yearly return the optimizer assumes an asset earns. Here it is the asset's average over the window, which is only an estimate.

GMV
Global minimum variance: the lowest-risk mix, whose value swung least over the window. Choosing it uses how the returns swing and move together, never their averages.

Hindsight
Knowing how things turned out. Weights chosen on the same prices they are scored on are chosen with hindsight.

Hold
In the walk-forward test, a stretch of days over which a mix is held unchanged, before its weights are chosen again.

In-sample
Scored on the same prices the weights were chosen from. It describes the past; it is not what anyone could have earned.

Index
A list of companies whose prices are followed together as one number, such as the S&P 500. A benchmark is an index, or a fund that tracks one.

Mega-cap
One of the very largest companies, measured by the total value of its shares.

Normal distribution
The bell curve. The page draws one beside the real returns to show where they differ from it.

Optimizer
The calculation that chooses weights to meet a goal, such as the least volatility or the highest Sharpe ratio.

Out-of-sample
Scored on days the weights had not seen when they were chosen. The fair test of a fitted mix.

Portfolio
A set of weights across the assets: a way of splitting the money. The page also calls one a mix.

Rebalanced
Bought and sold back to the target weights. Every mix on the page is rebalanced daily.

Return
How much an investment gained or lost, as a share of what was put in. A daily return is one day's change.

Risk contribution
An asset's share of a mix's risk, which can be far from its share of the money.

Risk-free rate
What a safe investment paid over the same time: short-term US Treasury bills. A Sharpe ratio counts only the return above it.

Rolling window
A fixed number of recent trading days that moves forward one day at a time, so a figure can be followed through time.

Short position
A negative weight: an asset sold after borrowing it, which gains when its price falls. Off unless switched on in the settings.

Skewness
Whether the biggest moves leaned toward gains (above zero) or toward losses (below zero).

Standard error
How far a figure estimated from a limited stretch of data could be off by chance alone. The page prints it as plus or minus.

Tangency
The best-tradeoff mix: the mix with the highest Sharpe ratio over the window, chosen with hindsight. Also called the maximum-Sharpe portfolio.

Ticker
The short code a market uses for an investment, such as AAPL for Apple.

Trading day
A day the market was open. Weekends and market holidays have no prices.

Volatility
How much returns swing about their average, as a yearly figure. The page's usual measure of risk.

Walk-forward test
Choosing weights on earlier prices, holding them over later ones, and repeating to the end: an out-of-sample test.

Weight
The share of a mix's money in one asset. A mix's weights add up to the whole.

What the i marks say

The page explains these figures in the small i mark beside each one, or under How it is calculated. Here is each in plain words; the page also gives it in finance terms and as the formula.

Alpha
The yearly return left over after allowing for how it moved with the benchmark. Above zero, it did better over this window than its beta explains; below zero, worse. It describes the past, not a skill that will last.

Annual return
How much money this investment made (or lost) per year on average, shown as a percentage.

Benchmark Sharpe
How much return the benchmark earned for each unit of risk over this window, on its own. No weights were chosen for it: it is there for comparison.

Best and worst month
The single best and worst calendar month over the window.

Beta
How much it tended to move when the benchmark moved: at one, in step with it; above one, further; below one, less.

Calmar ratio
Yearly growth divided by the worst fall: how much growth came with each point of the deepest loss.

Compound annual return
The steady yearly growth rate that turns the starting amount into the ending amount over this window.

Correlation
How closely the portfolio's daily moves follow the benchmark's, from −1 (opposite) to +1 (in step).

Cumulative return
How much the starting amount grew or shrank in total over the whole window.

Downside deviation
How much the returns swing, counting only the days that fell short of the risk-free rate.

Equal-weight Sharpe
How much return an even split of the money across every asset earned for each unit of risk over this window, with the split kept even. Nothing was fitted to pick these weights, so it is the yardstick the fitted mixes are measured against.

Expected shortfall
The average loss on the worst 5% of days in this window. Value at risk is the smallest of those losses; this is their average.

Fit window
All prior history refits on all the days before each hold, so the fit keeps growing. The other choice refits on only the most recent days, as many as the first fit had, so old years drop out.

GMV Sharpe (in-sample)
How much return the mix with the smallest day-to-day swings earned for each unit of risk over this window. The mix was picked on these same prices to swing as little as possible, not to score well, and the score is in-sample: it describes the past and promises nothing.

Hold length
How many trading days each set of weights is kept before the next refit. Shorter holds mean more refits, each on more recent prices.

Information ratio
The extra yearly return over the benchmark, divided by the tracking error: how far the portfolio's day-to-day returns strayed from the benchmark's.

Longest drawdown
The longest stretch spent below an earlier high before climbing back to it, or to the last day if it never did.

Maximum drawdown
The biggest drop from a high point to a low point — the worst loss you could have experienced.

Out-of-sample Sharpe ratio
The risk-vs-reward score earned on days the weights had not seen: each hold is scored with weights chosen before it began. The days of every hold are joined into one run and scored once.

Parameters estimated
How many numbers had to be estimated from the price history to choose these weights. Equal weight and a typed mix need none.

Positive months
The share of months that ended with a gain.

R squared
How much of the portfolio's daily movement the benchmark explains, from 0% to 100%.

Risk parity
A mix in which every asset adds the same share of the portfolio's day-to-day swings, so a calmer asset holds more money and a jumpier one less. It ignores average returns, but its weights come from this window's prices, so its figures are in-sample.

Sharpe ratio
A score for how much return came with the risk: the yearly return above the safe rate, for each unit of swing. Higher is better. It describes this window, not the years ahead.

Sharpe ratio and its standard error
The risk-vs-reward score, with a ± showing how far it could be off just from the luck of this window. A column whose weights were chosen on these same prices, as the line under its name says, could be off by more.

Sortino ratio
Like Sharpe, but it counts only the swings below the safe rate, not the swings up. Higher is better.

Standard error of the out-of-sample Sharpe ratio
How far the out-of-sample score could be off from the luck of these particular days alone. Two scores closer together than about one of these are well inside the noise; it is a scale for reading a gap, not a test of one.

Tangency on shrunk means
The maximum-Sharpe mix after every asset's average return has been pulled part of the way toward one shared figure, which narrows the gaps between them before the weights are chosen. How far they are pulled is worked out from the data; its figures are in-sample.

Tangency on the last year
The maximum-Sharpe mix chosen from the last year of prices only, then held at those weights over the whole window. Its figures are in-sample: the weights were picked with hindsight from part of these same prices.

Tangency Sharpe (in-sample)
The highest risk-vs-reward score any mix of these assets reached over this window, with the weights picked from the same prices it is scored on: an in-sample figure, not a forecast. The ± beside it is one standard error, how far the figure could miss by chance alone if the weights had been set before the window began. They were set on these same prices, so the true uncertainty is larger.

Tangency with a 25% cap
The maximum-Sharpe mix with no short positions and no asset above 25% of the money. Its weights were chosen on this window's prices, so its figures are in-sample.

The four other constructions
Adds the four other ways of choosing weights that the scorecard offers, each chosen again before every hold. One the basket or a fit cannot have is named, with the reason.

Tracking error
How much the portfolio's day-to-day returns differ from the benchmark's, scaled to a year.

Up and down capture
In months the benchmark rose (up) or fell (down), how much of its move the portfolio made, on average.

Value at risk
On the worst 5% of days in this window, the portfolio lost at least this much.

Volatility
How wildly the investment's value swings up and down. Higher = more unpredictable.

Weight lost to a one standard error cut
Lower the largest holding's expected return by one standard error, a change well inside its own noise, and solve again: how much of its weight it loses.

Weight range across lookbacks
Solve the portfolio again on the last 1, 2, 3 and 5 years and on the full window: the most any one asset's weight moves between those answers.

Weight range across redraws
Draw expected returns anywhere their own noise allows, many times, and solve on each: the range the largest holding's weight covers, 10th to 90th percentile.

How the figures are computed

Every formula and data source is listed under **Methodology** at the foot of the page, and of this guide. The code is public, linked beside it.

Methodology

Prices
Daily adjusted closes from Yahoo Finance (adjusted for splits and dividends), for each ticker and the benchmark.

Cleaning
A ticker missing more than 5% of the trading days in the range is dropped (the benchmark never is); the rest are kept only on days when every series has a price. The range must span at least 2 years, and at least 3 tickers and 252 days of prices must remain.

Returns
Daily simple returns: each day's price over the day before, minus one.

Annualisation
Mean daily return × 252; standard deviation × √252.

Risk-free rate
The mean 3-month Treasury yield from FRED (series DGS3MO) over the days the prices cover, with the latest yield shown beside it in the settings; or the rate typed there; the example shown before live prices arrive keeps the rate it was saved with; 2% when FRED cannot be reached.

Sharpe ratio
(Annual return − risk-free rate) ÷ annual volatility.

Sortino ratio
(Annual return − risk-free rate) ÷ downside deviation: the root-mean-square shortfall of daily returns below the daily risk-free rate, taken over every day (days above it count as zero), × √252.

Skewness and kurtosis
The bias-corrected sample statistics; kurtosis is excess kurtosis, so a normal distribution scores 0.

Maximum drawdown
The largest fall from a running peak in the value of $1 invested on the first day.

Portfolios over time
Fixed weights applied to every day's returns: each portfolio is rebalanced daily to its target weights.

Portfolio volatility
$\sqrt{(w^T \Sigma w)}$, from the covariance matrix of daily returns, annualised.

Optimisation
The minimum-variance (GMV) and maximum-Sharpe (tangency) portfolios are solved exactly, as quadratic programmes (the Goldfarb–Idnani method, via quadprog). Weights sum to 100%; each lies in [0%, 100%], or in [−100%, 100%] with short positions allowed. When no portfolio earns more than the risk-free rate, the page says so.

Efficient frontier
The lowest volatility reachable at 80 target returns, from the GMV portfolio's return up to the highest return the bounds allow, each an exact quadratic programme. The capital allocation line runs from the risk-free rate through the tangency portfolio.

Risk contribution
Each asset's share of portfolio variance, $w_i(\Sigma w)_i \div w^T \Sigma w$; the shares sum to 100%.

CAPM beta and alpha
An ordinary least-squares regression of the asset's daily excess return on the benchmark's; beta is the slope, alpha the intercept × 252.

Rolling figures
Volatility and correlation over trailing windows of 30, 60, 90 or 120 trading days (60 to start).

Estimation windows
Trailing 1, 2, 3 and 5 years (252 trading days a year) and the full sample, each optimised and scored on its own returns: in-sample, a description of the fit rather than of what the weights went on to earn.

Custom weights
Each weight is held to the bounds, then divided by their total. With short positions allowed, a net exposure of 5% or less, or a weight that normalising pushes past ±100%, is refused with a message.

Choosing the assets

The published test's five mega-caps were picked for being large today, which is hindsight in the choice of assets: it inflates the absolute Sharpe ratios, and it applies to all three portfolios alike.

Literature

The walk-forward test repeats DeMiguel, Garlappi and Uppal (2009), *Optimal Versus Naive Diversification: How Inefficient Is the 1/N Portfolio Strategy?*, Review of Financial Studies 22(5). The redraws and what-ifs on expected returns show the fragility described by Michaud (1989), *The Markowitz Optimization Enigma: Is 'Optimized' Optimal?*, Financial Analysts Journal 45(1). The counter-argument is Kritzman, Page and Turkington (2010), *In Defense of Optimization: The Fallacy of 1/N*, Financial Analysts Journal 66(2): optimisation beats an equal split once its expected returns are not sample means from short windows. The Sharpe ratio's standard error is Lo (2002), *The Statistics of Sharpe Ratios*, Financial Analysts Journal 58(4), with Mertens' (2002) correction; the shrunk-mean tangency is Jorion (1986), *Bayes-Stein Estimation for Portfolio Analysis*, Journal of Financial and Quantitative Analysis 21(3).

Origin

Began as a graduate course project and was improved afterwards.

Returns are daily simple returns from adjusted closing prices, annualised over 252 trading days. Portfolio weights sum to 100%; each lies between 0% and 100%, or between −100% and 100% with short positions allowed. Every figure describes the period shown and is not a forecast.

Source code on GitHub · masonjbennett.com